Question Paper

Exam Date & Time: 02-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

MATHEMATICS FOR FINANCE [MAT 2132]

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) (4)

A) Find the LU factorization of the matrix $A=\begin{bmatrix}3 & 1 & 2\\ -9 & 0 & -4\\ 9 & 9 & 14\end{bmatrix}$

B) Apply the Gram-Schmidt algorithm to orthogonalize the basis{(1,1,1), (1,0,1), (1,1,2)} (3)

C) Determine the definiteness of the quadratic form $x^2+2y^2-7z^2-4xy+8xz$ (3)

2) (4)

A) Find the least squares solution of $Ax=b$, where $A=\begin{bmatrix}0 & 1\\ 1 & 1\\ 2 & 1\end{bmatrix}$ and $b=\begin{bmatrix}6\\ 0\\ 0\end{bmatrix}$. What quantity is being minimized?

B) Find Nul A where $A=\begin{bmatrix}3 & 6 & 6 & 3 & 9\\ 6 & 12 & 13 & 0 & 3\end{bmatrix}$ (3)

C) Determine whether $u=\begin{bmatrix}3\\ 3\\ 8\end{bmatrix}$ and $v=\begin{bmatrix}-2\\ 5\\ 1\end{bmatrix}$ are in the column space of $A=\begin{bmatrix}2 & 5 & 1\\ -1 & -7 & -5\\ 3 & 4 & -2\end{bmatrix}$ (3)

3) (4)

Find the matrix that diagonalizes the matrix $A=\begin{bmatrix}2 & 6\\ 0 & 1\end{bmatrix}$

A)

B) (3)

Find the singular value decomposition of the matrix $A=\begin{bmatrix}3 & 0\\ 4 & 5\end{bmatrix}$

C) Suppose that a savings account initially contains Rupees 40 and earns 6% interest compounded annually. Determine a formula which describes how to compute each year's balance from the previous year's balance. Also, solve the obtained equation (3)

4) Solve $y^{"}+(1+x)y'-y=0$, $y(0)=y'(0)$, $y(1)+y'(1)=1$, with $h=0.5$ (4)

A)

B) (3)

Find the solution of the equation $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,0)=0$, $u(0,t)=0$

and $u(1,t) = t$. Use Crank-Nicolson scheme. Take $k = 1/8$ and $h = 1/4$.

C) (3)

Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, |x| \leq 1 \\ 0, |x| > 1 \end{cases}$. Hence evaluate $\int_0^\infty \frac{x\cos x - \sin x}{x^3} \cos\frac{x}{2}\, dx.$

5) (3)

Solve $u_{xx} + u_{yy} = 0 \quad 0 < x < 1 \quad 0 < y < 1$ in , , given the boundary conditions

A)

$u(x,1) = u(0,y) = 0 \quad u(x,0) = 9(x - x^2) \quad u(1,y) = 9(y - y^2) \qquad h = 1/3$

, , . Take .

Use Gauss-Seidel method for solving. Perform two iterations

B) (4)

Find a Fourier series expansion for $f(x) = \begin{cases} -\pi, -\pi < x < 0 \\ x, 0 < x < \pi \end{cases}$. Deduce that

$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$

C) The turning moment T is given for a series of values of the crank angle θ°. Obtain the first four terms in a series of sines to represent T. Also, obtain T for $\theta = 75^\circ$ (3)

θ°	0	30	60	90	120	150	180
T	0	5224	8097	7850	5499	2626	0

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